

RAJENDRA PRASAD N

LEAD SRE | AI, GPU/HPC, KUBERNETES & OBSERVABILITY

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Lead Site Reliability and Observability Engineer with more than a decade of experience building and operating cloud, Kubernetes, and high-availability production platforms. Specialized in GPU-enabled AI infrastructure, high-performance computing, large-scale Kubernetes, data-centre systems, and open-source observability. Hands-on across Prometheus, Grafana, VictoriaMetrics, Loki, Thanos, OpenTelemetry, Terraform, Python, Linux, AWS, EKS, Helm, Milvus, and distributed telemetry architectures. Proven in environments that cannot tolerate downtime, with deep focus on metrics, logs, traces, alerting, SLOs, incident response, capacity planning, performance engineering, automation, and reliable operation of critical AI workloads.

CERTIFICATIONS



GitLab Certified Associate



Microsoft Certified: Azure Fundamentals



Microsoft Certified: Azure DevOps Engineer Expert



AWS Certified Solutions Architect - Associate



MuleSoft Certified Developer - Level 1

SRE, GPU/HPC & OBSERVABILITY HIGHLIGHTS

- GPU & HPC Platforms** Engineered Kubernetes-based GPU compute platforms for AI and high-performance workloads, with capacity, scheduling, reliability, and utilization controls.
- Observability at Scale** Built open-source metrics, logs, traces, dashboards, and alerting platforms using Prometheus, Grafana, Loki, Thanos, VictoriaMetrics, and OpenTelemetry.
- Site Reliability** Applied SLOs, error budgets, incident response, capacity planning, automation, and failure-mode analysis to services with extremely low downtime tolerance.
- Kubernetes Engineering** Operated large-scale Kubernetes and EKS environments with Helm, GitOps, workload isolation, autoscaling, policy controls, and production-grade monitoring.
- Infrastructure as Code** Developed reusable Terraform modules and automated provisioning patterns for compute, networking, storage, observability, and Kubernetes infrastructure.

CORE TECHNOLOGIES

GPU & HPC

GPU compute, high-performance computing, AI/ML platforms, GPU node pools, workload scheduling, capacity planning, performance telemetry

OBSERVABILITY

Prometheus, Grafana, VictoriaMetrics, Loki, Thanos, OpenTelemetry, Alertmanager, metrics, logs, traces, alerting

SITE RELIABILITY ENGINEERING

SLIs, SLOs, error budgets, incident response, root-cause analysis, toil reduction, capacity management, disaster recovery

KUBERNETES

Kubernetes, EKS, OpenShift, Helm, Argo CD, operators, autoscaling, resource quotas, network policies, multi-cluster operations

INFRASTRUCTURE AS CODE

Terraform, reusable modules, remote state, state locking, CloudFormation, Ansible, Packer, policy validation

CLOUD & DATA CENTRES

AWS, hybrid cloud, data-centre infrastructure, Linux, compute, storage, networking, private connectivity, availability zones

AI DATA PLATFORMS

Milvus, vector search, Databricks, GPU-enabled workloads, distributed data services, AI platform reliability

TELEMETRY ENGINEERING

PromQL, LogQL, dashboards, recording rules, alert rules, exporters, service instrumentation, cardinality management

AUTOMATION

Python, Bash, PowerShell, YAML, JSON, APIs, CI/CD, GitOps, operational tooling

DEVSECOPS

IAM, secrets management, container scanning, policy controls, least privilege, auditability, secure platform operations

PROFESSIONAL EXPERIENCE

Lead SRE - AI, GPU & Observability | AT&T Services Inc.

08/2023 - Present

Skills Used: GPU/HPC, Kubernetes, AWS, Terraform, Prometheus, Grafana, Loki, OpenTelemetry, Milvus, Python

- Designed and operated Kubernetes-based GPU compute platforms for AI workloads, integrating GPU node pools, workload isolation, scheduling controls, resource quotas, autoscaling, and capacity planning.
- Built high-throughput vector-search infrastructure with Milvus and GPU-enabled services, improving scalability and production readiness for AI-oriented application platforms.
- Owned the observability architecture for large-scale Kubernetes workloads across metrics, logs, traces, dashboards, and alerting using Prometheus, Grafana, Loki, Thanos, VictoriaMetrics, and OpenTelemetry patterns.
- Implemented service and platform telemetry with OpenTelemetry collectors, exporters, consistent resource attributes, sampling controls, and trace-to-log and trace-to-metric correlation.
- Designed scalable metrics pipelines with recording rules, long-term retention, federation, cardinality controls, and highly available query paths for infrastructure and application telemetry.
- Established SLOs, SLIs, error budgets, multi-window alerting, escalation policies, and operational dashboards for platforms where downtime directly affected critical workloads.
- Developed reusable Terraform modules to provision GPU compute, Kubernetes, networking, storage, identity, and observability components through repeatable infrastructure-as-code workflows.
- Automated monitoring configuration, alert validation, incident diagnostics, capacity reporting, and platform recovery workflows with Python and Bash to reduce operational toil.
- Led incident response and root-cause analysis for Kubernetes, GPU scheduling, telemetry ingestion, storage, networking, and infrastructure failures, converting findings into durable automation and runbooks.
- Partnered with AI, platform, data, network, storage, and data-centre teams to forecast demand, eliminate bottlenecks, and improve GPU utilization, workload throughput, and service reliability.

Senior SRE - Kubernetes Observability | GlobalLogic

03/2023 - 08/2023

Skills Used: Kubernetes, Terraform, Prometheus, Grafana, VictoriaMetrics, Loki, Thanos, OpenTelemetry

- Engineered highly available open-source observability services for Kubernetes, including metrics ingestion, log aggregation, distributed tracing, dashboards, and actionable alerting.
- Standardized Prometheus rules, Grafana dashboards, Loki queries, OpenTelemetry instrumentation, and alert routing across application and platform teams.
- Tuned retention, compaction, sharding, caching, replication, and query performance to support high-volume telemetry without compromising platform reliability.
- Deployed observability components through Helm and Terraform and applied GitOps controls for repeatable configuration, promotion, rollback, and disaster recovery.
- Diagnosed Kubernetes resource pressure, noisy-neighbor behavior, network latency, storage saturation, and telemetry pipeline failures using correlated metrics, logs, and traces.

Senior DevOps / SRE Engineer | Akamai Technologies

12/2021 - 03/2023

Skills Used: AWS, Kubernetes, Terraform, Prometheus, Grafana, Python, Linux

- Operated distributed Kubernetes services across cloud and hybrid infrastructure with production monitoring, autoscaling, deployment automation, and recovery controls.
- Provisioned infrastructure through Terraform and pipeline-driven IaC patterns for consistent multi-environment delivery.
- Implemented custom exporters and telemetry automation in Python to expose application, infrastructure, and workload-specific operational metrics.
- Built dashboards and alerts for saturation, latency, errors, availability, queue depth, capacity, and deployment health.
- Troubleshoot complex compute, container, network, storage, and release issues across distributed engineering teams.

Senior Site Reliability Engineer | Turvo India Pvt Ltd

04/2020 - 12/2021

Skills Used: Kubernetes, Prometheus, Grafana, Terraform, Python, Ansible, Linux

- Developed custom Prometheus exporters in Python to capture application-specific operational metrics for SRE use cases.
- Built resilient automation to deploy highly available Prometheus, Grafana, and node-level monitoring across shared platforms.
- Defined reliability indicators and operational dashboards for availability, latency, traffic, errors, saturation, and resource utilization.
- Reduced operational toil through automated provisioning, configuration enforcement, health checks, and recovery workflows.
- Applied capacity planning and performance analysis to maintain reliability during workload growth and infrastructure change.

Senior DevOps Engineer - Data Centre Platforms | HCL Technologies

02/2019 - 04/2020

Client: Mercedes-Benz Research & Development India

Skills Used: Linux, Kubernetes, VMware, Terraform, Ansible, ELK, Prometheus

- Supported Linux, virtualization, compute, networking, storage, and Kubernetes services across enterprise data-centre environments.
- Designed monitoring architecture for application, infrastructure, and platform telemetry using open-source collectors and the ELK stack.
- Deployed Kubernetes workloads with Helm and standardized release packaging across multi-namespace environments.
- Automated infrastructure configuration and operational controls with Terraform, Ansible, shell scripting, and PowerShell.
- Created production runbooks and supported high-availability releases, incident response, and service restoration.

DevOps / Platform Engineer | Mphasis Ltd

08/2016 - 12/2018

Client: HPE

Skills Used: Linux, Terraform, Ansible, Python, Shell, Splunk, Kubernetes

- Authored reusable Terraform modules and automated infrastructure changes through governed CI/CD workflows.
- Built platform monitoring, log analysis, and operational dashboards for distributed application environments.
- Automated configuration, deployment, health checks, and recovery tasks using Python, shell scripting, and Ansible.
- Supported self-hosted execution infrastructure and diagnosed compute, dependency, networking, and environment failures.
- Improved platform reliability through repeatable deployments, controlled promotion, documentation, and operational standardization.

EDUCATION

B.Tech (2009-2013) in Information Technology
Prakasam Engineering College, Jawaharlal Nehru Technological University, Kakinada